

Nirma University

Institute of Technology

Semester End Examination (IR), January-2022

MCA Semester-1

3MCA105 Computer Networks

Roll/
Exam No.

Supervisor's Initial
with date

Time: 2 Hours

Max Marks: 50

Instructions:

1. Attempt all questions.
2. Figures to the right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.

- Q-1** Explain the key characteristics of Personal Area Network (PAN), [8]
CO 1 BL 3 Local Area Network (LAN), Metropolitan Area Network (MAN) and Wide Area Network (WAN).
- Q-2** For a domain 192.165.20.0, find the class, the number of bits in the [8]
CO 1,2,3 BL 3 host id, the number of IP addresses, the number of usable hosts, the subnet mask, directed broadcast address, first usable host address and last usable host address in this domain.
- Q-3** For given data-word 0011, generate code-word at sender site using [8]
CO 1,2 BL 3 hamming code error detection and correction method. Show step by step procedure for generating code-word. The third most significant bit gets inverted in transmission. Show step by step procedure to detect and correct error at receiver site.
- Q-4** Explain the TCP connection termination process and illustrate the [8]
CO 1,2 BL 4 same with the help of time-line diagram.
- OR**
- Q-4** How Selective-Repeat ARQ protocol provides *error control* and *flow* [8]
CO 1,2 BL 4 *control*? Illustrate it by taking suitable example.
- Q-5** Explain usefulness of Network Address Translation technique. Also [7]
CO 1,2 BL 3 explain with the help of example how does it work.
- OR**
- Q-5** Explain how does File Transfer Protocol work? Discuss any one [7]
CO 1,2 BL 3 drawback of it.
- Q-6** Critically differentiate between 1-persistent and non-persistent [4]
CO 1,2 BL 3 method.
- Q-7** Why there is restriction on frame size in CSMA/CD? Justify your [4]
CO 1,2 BL 5 answer by giving suitable example.
- Q-8** A non-periodic composite signal has a bandwidth of 100 kHz, with [3]
CO 1,2 BL 3 a middle frequency of 150 kHz and peak amplitude of 10 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.